

导数

1. $(C)' = 0,$	2. $(C \cdot f(x))' = C f'(x),$	$(\sin x)' = \cos x,$	$(\cos x)' = -\sin x,$	$(\tan x)' = \sec^2 x,$
3. $\left(\sum_{k=1}^n f_k\right)' = \sum_{k=1}^n f_k',$	4. $\left(\prod_{k=1}^n f_k\right)' = \sum_{k=1}^n \left(f_k' \prod_{j=1, j \neq k}^n f_j\right),$	$(\csc x)' = -\cot x \csc x,$	$(\sec x)' = \tan x \sec x,$	$(\cot x)' = -\csc^2 x,$
5. $\left(\frac{f}{g}\right)' = \frac{f'g - g'f}{g^2},$	6. $(f(\varphi(x)))' = f'(\varphi)\varphi'(x),$	$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}},$	$(\arccos x)' = -\frac{1}{\sqrt{1-x^2}},$	$(\arctan x)' = \frac{1}{1+x^2},$
7. $(a^x)' = a^x \ln a, (e^x)' = e^x,$	8. $(\log_a x)' = \frac{1}{x \ln a}, (\ln x)' = \frac{1}{x},$	$(\operatorname{arccsc} x)' = \frac{-1}{x\sqrt{x^2-1}},$	$(\operatorname{arcsec} x)' = \frac{1}{x\sqrt{x^2-1}},$	$(\operatorname{arccot} x)' = \frac{-1}{1+x^2},$
9. $(x^a)' = ax^{a-1},$	10. $(\ln x + \sqrt{x^2 \pm b})' = \frac{1}{\sqrt{x^2 \pm b}},$	$(\sinh x)' = \cosh x,$	$(\cosh x)' = \sinh x,$	$(\tanh x)' = \operatorname{sech}^2 x,$
		$(\operatorname{csch} x)' = -\operatorname{csch} x \coth x,$	$(\operatorname{sech} x)' = -\operatorname{sech} x \tanh x,$	$(\coth x)' = -\operatorname{csch}^2 x,$
		$(\operatorname{arsinh} x)' = \frac{1}{\sqrt{x^2+1}},$	$(\operatorname{arcosh} x)' = \frac{1}{\sqrt{x^2-1}},$	$(\operatorname{artanh} x)' = \frac{1}{1-x^2},$
		$(\operatorname{arcsch} x)' = \frac{-1}{x\sqrt{1+x^2}},$	$(\operatorname{arsech} x)' = \frac{-1}{x\sqrt{1-x^2}},$	$(\operatorname{arcoth} x)' = \frac{-1}{x^2-1},$

常用高阶导数公式

熟记!

$(1) (a^x)^{(n)} = a^x \cdot \ln^n a \quad (a > 0) \quad (e^x)^{(n)} = e^x$
 $(2) (\sin kx)^{(n)} = k^n \sin(kx + n \cdot \frac{\pi}{2})$
 $(3) (\cos kx)^{(n)} = k^n \cos(kx + n \cdot \frac{\pi}{2})$
 $(4) (x^\alpha)^{(n)} = \alpha(\alpha-1) \cdots (\alpha-n+1)x^{\alpha-n}$
 $(5) (\ln x)^{(n)} = (-1)^{n-1} \frac{(n-1)!}{x^n} \quad (\frac{1}{x})^{(n)} = (-1)^n \frac{n!}{x^{n+1}}$
 $(\frac{1}{x+a})^{(n)} = (-1)^n \frac{n!}{(x+a)^{n+1}}$

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$$\begin{aligned} f(x) + f'(x) &= e^x f(x) \\ f'(x) + x f(x) &= e^{\frac{x^2}{2}} f(x) \\ \frac{1}{x} f(x) + \ln x f'(x) &= \ln x f(x) \end{aligned}$$

$p_n(x)$ 是 n 次多项式, 导一次, 次数少一次
 n 阶导 $\rightarrow$ 常数, 阶数高于 n , 导数 $= 0$

$$\begin{aligned} (\log_a x)^{(n)} &= (-1)^{n-1} \frac{(n-1)!}{x^n \ln a}, \quad n > 0 \text{ 且 } a \neq 1 \\ (\frac{1}{a-x})^{(n)} &= \frac{n!}{(a-x)^{n+1}} \end{aligned}$$

$$\frac{d^2 y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}} = \frac{dy'}{dt}$$

泰勒公式

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \frac{e^\theta x^{n+1}}{(n+1)!} \quad (0 < \theta < 1)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + o(x^n)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + (-1)^{n-1} \frac{x^n}{n} + o(x^n)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + o(x^{2n+2})$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + o(x^{2n+1})$$

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \cdots + \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{n!} x^n + o(x^n)$$

$$\tan x = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + o(x^5)$$

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + R_n(x)$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots + x^n + R_n(x)$$

$$\alpha = \frac{1}{2}, \quad \sqrt{1+x} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + R_2(x)$$

$$\alpha = \frac{1}{3}, \quad (1+x)^{\frac{1}{3}} = 1 + \frac{1}{3}x - \frac{1}{9}x^2 + R_2(x)$$

$$(1) \int 0 dx = C; \int \frac{1}{x} dx = -\frac{1}{x} + C \quad \int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + C$$

$$(2) \int x^a dx = \frac{1}{a+1} x^{a+1} + C \quad (a \neq -1);$$

$$(3) \int \frac{1}{x} dx = \ln|x| + C;$$

$$(4) \int e^x dx = e^x + C;$$

$$(5) \int a^x dx = \frac{a^x}{\ln a} + C \quad (a > 0, a \neq 1); \rightarrow \frac{\sinh x}{\cosh x}$$

$$(6) \int \cos x dx = \sin x + C; \int \tan x dx = -\ln|\cos x| + C$$

$$(7) \int \sin x dx = -\cos x + C; \int \cot x dx = \ln|\sin x| + C$$

$$(8) \int \frac{1}{\cos^2 x} dx = \int \sec^2 x dx = \tan x + C;$$

$$(9) \int \frac{1}{\sin^2 x} dx = \int \csc^2 x dx = -\cot x + C;$$

$$(10) \int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C \quad (\text{或 } -\arccos x + C_1);$$

$$\int \sec x \tan x dx = \sec x + C \quad \int \csc x dx = \ln|\csc x - \cot x| + C$$

$$\int \csc x \cot x dx = -\csc x + C \quad \int \sec x dx = \ln|\sec x + \tan x| + C$$

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$$(11) \int \frac{1}{1+x^2} dx = \arctan x + C \quad (\text{或 } -\operatorname{arccot} x + C_1);$$

$$(12) \int \cosh x dx = \sinh x + C;$$

$$(13) \int \sinh x dx = \cosh x + C;$$

$$(14) \int \frac{1}{\sqrt{x^2+1}} dx = \operatorname{arsh} x + C = \ln(x + \sqrt{x^2+1}) + C;$$

$$(15) \int \frac{1}{\sqrt{x^2-1}} dx = \operatorname{arch} x + C = \ln(x + \sqrt{x^2-1}) + C.$$

$$\int \frac{1}{a^2+x^2} = \frac{1}{a} \arctan \frac{x}{a} + C \quad \int \tan x dx = -\ln|\cos x| + C$$

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C \quad \int \cot x dx$$

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin \frac{x}{a} + C \quad \int \csc x dx$$

$$\int \frac{dx}{a^2-x^2} = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$$

$$\int \frac{1}{\sqrt{x^2 \pm a^2}} dx = \ln|x + \sqrt{x^2 \pm a^2}| + C$$

$$(1) x^{n-1} f(kx^n + p) dx = \frac{1}{kn} f(kx^n + p) d(kx^n + p);$$

$$(2) e^x f(ke^x + p) dx = \frac{1}{k} f(ke^x + p) d(ke^x + p),$$

$$a^x f(ka^x + p) dx = \frac{1}{k \ln a} f(ka^x + p) d(ka^x + p);$$

$$(3) \cos x f(k \sin x + p) dx = \frac{1}{k} f(k \sin x + p) d(k \sin x + p),$$

$$\sin x f(k \cos x + p) dx = -\frac{1}{k} f(k \cos x + p) d(k \cos x + p),$$

$$\frac{1}{\cos^2 x} f(k \tan x + p) dx = \frac{1}{k} f(k \tan x + p) d(k \tan x + p);$$

$$(4) \frac{1}{x} f(k \ln x + p) dx = \frac{1}{k} f(k \ln x + p) d(k \ln x + p),$$

$$\frac{1}{x} f(k \log_a x + p) dx = \frac{\ln a}{k} f(k \log_a x + p) d(k \log_a x + p);$$

$$(5) f(\arcsin x) \cdot \frac{dx}{\sqrt{1-x^2}} = f(\arcsin x) d(\arcsin x),$$

$$f(\arctan \frac{x}{a}) \cdot \frac{dx}{a^2+x^2} = \frac{1}{a} f(\arctan \frac{x}{a}) d(\arctan \frac{x}{a})$$

$$(6) \frac{x dx}{1+x^2} = \frac{1}{2} d \ln(1+x^2).$$

$$\frac{1}{\sin x} \textcircled{1} \frac{1}{2 \sin \frac{x}{2} \cos \frac{x}{2}}$$

$$\textcircled{2} \int \frac{\sin x}{\sin^2 x} dx = \int \frac{d(\cos x)}{1-\cos^2 x}$$

点评 能否熟练地凑微分,是求不定积分的重要技巧之一,

$$(1) x^{n-1} f(kx^n + p) dx = \frac{1}{kn} f(kx^n + p) d(kx^n + p);$$

$$(2) e^x f(ke^x + p) dx = \frac{1}{k} f(ke^x + p) d(ke^x + p),$$

$$a^x f(ka^x + p) dx = \frac{1}{k \ln a} f(ka^x + p) d(ka^x + p);$$

$$(3) \cos x f(k \sin x + p) dx = \frac{1}{k} f(k \sin x + p) d(k \sin x + p),$$

$$\sin x f(k \cos x + p) dx = -\frac{1}{k} f(k \cos x + p) d(k \cos x + p),$$

$$\frac{1}{\cos^2 x} f(k \tan x + p) dx = \frac{1}{k} f(k \tan x + p) d(k \tan x + p);$$

$$(4) \frac{1}{x} f(k \ln x + p) dx = \frac{1}{k} f(k \ln x + p) d(k \ln x + p),$$

$$\frac{1}{x} f(k \log_a x + p) dx = \frac{\ln a}{k} f(k \log_a x + p) d(k \log_a x + p);$$

$$(5) f(\arcsin x) \cdot \frac{dx}{\sqrt{1-x^2}} = f(\arcsin x) d(\arcsin x),$$

$$f(\arctan \frac{x}{a}) \cdot \frac{dx}{a^2+x^2} = \frac{1}{a} f(\arctan \frac{x}{a}) d(\arctan \frac{x}{a});$$

$$(6) \frac{x dx}{1+x^2} = \frac{1}{2} d \ln(1+x^2).$$

$$\int \sqrt{x^2 \pm a^2} dx = \frac{x}{2} \sqrt{x^2 \pm a^2} \pm \frac{a^2}{2} \ln|x + \sqrt{x^2 \pm a^2}| + C$$

$$\tan x - \sin x = \tan x (1 - \cos x)$$

$$\arctan 1 = \frac{\pi}{4}$$

线密度 (微分) 功/曲边梯形 (积分)

$$d = \max\{\Delta b\} \quad d \rightarrow 0$$

$\int_a^b V(b)$ 定积分 $\rightarrow$ 大化小, 常代变, 近似和, 取极限

匀 $\leftarrow$ 非匀 (极限)

遇到绝对值分类讨论

看清/看全题 x 从低到高

$$\ln 8 = 3 \ln 2$$

$$e^{x \cos x^2} = e^x e^{x(\cos x^2 - x)}$$

$$\lim_{x \rightarrow 0} x^2 \ln \frac{1}{x} \quad (\text{不能用等价无穷小})$$

有界

$$(1+\Delta)^n - 1 = \alpha \Delta$$

Δ 和差化积公式

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha+\beta}{2} \cos \frac{\alpha-\beta}{2}$$

$$\sin \alpha - \sin \beta = 2 \cos \frac{\alpha+\beta}{2} \sin \frac{\alpha-\beta}{2}$$

$$\cos \alpha + \cos \beta = 2 \cos \frac{\alpha+\beta}{2} \cos \frac{\alpha-\beta}{2}$$

$$\cos \alpha - \cos \beta = -2 \sin \frac{\alpha+\beta}{2} \sin \frac{\alpha-\beta}{2}$$

只可对乘积因子作无穷小变换

和一般不可以

$$\beta = \alpha + o(\alpha)$$

$$\sec^2 x = \tan^2 x + 1$$

积化和差

$$\sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha+\beta) + \sin(\alpha-\beta)]$$

$$\cos \alpha \sin \beta = \frac{1}{2} [\sin(\alpha+\beta) - \sin(\alpha-\beta)]$$

$$\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha+\beta) + \cos(\alpha-\beta)]$$

$$\sin \alpha \sin \beta = -\frac{1}{2} [\cos(\alpha+\beta) - \cos(\alpha-\beta)]$$

极限

$$\lim_{x \rightarrow x_0} f(x) = A \Leftrightarrow \forall \epsilon > 0, \exists \delta > 0, \text{当 } 0 < |x - x_0| < \delta \text{ 时, 有 } |f(x) - A| < \epsilon.$$

$$\lim_{x \rightarrow +\infty} f(x) = A \Leftrightarrow \forall \epsilon > 0, \exists X > 0, \text{当 } |x| > X \text{ 时, 有 } |f(x) - A| < \epsilon.$$

$$\text{设 } a > 0, \lim_{n \rightarrow \infty} \sqrt[n]{a} = 1$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{n} = \begin{cases} 0, & |q| < 1 \\ \infty, & |q| > 1 \end{cases}$$

$$\lim_{x \rightarrow 0^+} x^x = 1 \quad \lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$$

$$\text{任何 } a > 0, \lim_{x \rightarrow \infty} \frac{x^a}{e^x} = 0, \lim_{x \rightarrow 0^+} x^a \ln x = 0$$

$$a^x \quad [a \in (0, 1) \quad a = 1, a \in (1, \infty)]$$

$$\lim_{x \rightarrow \infty} \arctan x = \frac{\pi}{2}$$

$$\lim_{x \rightarrow 0} (1 + \frac{a}{x})^x = e^a, \quad \lim_{x \rightarrow 0} (1 + ax)^{\frac{1}{x}} = e^a$$

$$R(x) = \frac{a_0 x^n + a_1 x^{n-1} + \dots + a_n}{b_0 x^m + b_1 x^{m-1} + \dots + b_m}$$

$$\lim_{x \rightarrow \infty} R(x) = \begin{cases} \infty, & n > m \\ \frac{a_0}{b_0}, & n = m \quad (\text{分子、分母同除 } x^m) \\ 0, & n < m \end{cases}$$

当 $x \rightarrow 0$ 时

$$\sin x \sim x$$

$$\ln(1+x) \sim x$$

$$\arcsin x \sim x$$

$$\log_a(1+x) \sim \frac{x}{\ln a}$$

$$1 - \cos x \sim \frac{1}{2} x^2$$

$$x - \sin x \sim \frac{1}{6} x^3$$

$$(1+x)^a - 1 \sim ax$$

$$x - \arctan x \sim \frac{1}{3} x^3$$

$$\tan x \sim x$$

$$e^x - 1 \sim x$$

$$\arctan x \sim x$$

$$a^x - 1 \sim x \ln a$$

$$\sqrt[n]{1+x} - 1 \sim \frac{x}{n}$$

$$\tan x - x \sim \frac{1}{3} x^3$$

$$\arcsin x - x \sim \frac{1}{6} x^3$$

$$\tan x - \sin x \sim \frac{1}{2} x^3$$

$$\tan x - \sin x \sim \frac{1}{2} x^3$$

$$\lim_{x \rightarrow 0} e^{\frac{1}{x}}, \lim_{x \rightarrow 0} \ln \frac{1}{x}, \lim_{x \rightarrow 0} \cos \frac{1}{x}, \lim_{x \rightarrow \infty} \sin x, \lim_{x \rightarrow \infty} \cos x$$

$\rightarrow$ 极限不存在也不为 ∞

$$\text{若 } a > 1, \text{ 则 } \lim_{x \rightarrow +\infty} a^x = +\infty, \lim_{x \rightarrow -\infty} a^x = 0.$$

$$0 < a < 1, \lim_{x \rightarrow +\infty} a^x = 0, \lim_{x \rightarrow -\infty} a^x = +\infty$$

$$a > 0, \lim_{n \rightarrow \infty} \sqrt[n]{a} = 1$$

等价无穷小

导数在实际中应用

$$V = \frac{1}{3} \pi r^2 h$$

与h有关, 两边求导

广义积分中值

$f(x)$ 在 $[a, b]$ 上连续, $g(x)$ 在 $[a, b]$ 可积, 不变号

$[a, b]$ 上存在一点

$$\int_a^b f(x) g(x) dx = f(\xi) \int_a^b g(x) dx$$

微积分基本公式

设 $f(x) \in C[a, b]$, 且 $F'(x) = f(x)$, 则有 $\xi \in (a, b)$

$$\int_a^b f(x) dx = f(\xi)(b-a) = F'(\xi)(b-a) = F(b) - F(a)$$

积分 微分

求微积分的方法

三角有理化
有理化
无理式
凑微分

函数
极限
连续

间断 $\begin{cases} \text{第一类 (极限存在)} \\ \text{第二类} \end{cases} \begin{cases} \text{跳跃} \\ \text{可去} \\ \text{无穷} \\ \text{振荡} \end{cases}$

证明连续. 求 a 处 $f(a)$, $\lim_{x \rightarrow a} f(x)$ 2 相同

$\varphi(x) = 0$ $\begin{cases} \text{零点} \\ \text{将 } \varphi(x) \text{ 看作导数, Roll 定理} \end{cases} \rightarrow \text{求原函数}$

Roll

导数
微分

$$dy = f'(x) dx$$

中值定理 $\begin{cases} f'(\xi) = 0 \\ f(\xi) = \frac{f(b) - f(a)}{b - a} \end{cases} \quad \begin{matrix} \theta \in (0, 1) \\ f(x_0 + \theta \Delta x) - f(x_0) = f'(x_0 + \theta \Delta x) \Delta x \\ \text{增量公式} \end{matrix}$

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}$$

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \dots + \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n + R_n(x)$$

其中 $R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1}$ (ξ 在 x_0 与 x 之间)

$$R = \frac{1}{K}$$

$$K = \frac{|y''|}{(1 + (y')^2)^{\frac{3}{2}}}$$

$$\begin{cases} \xi = x - \frac{y'(1 + (y')^2)}{y''}, \\ \eta = y + \frac{1 + (y')^2}{y''}. \end{cases}$$

$$(x - \xi)^2 + (y - \eta)^2 = R^2$$

$$ds = \sqrt{(dx)^2 + (dy)^2} \quad \text{弧微分}$$

$$ds = \sqrt{(x'(t))^2 + (y'(t))^2} dt$$

$$ds = \sqrt{(r'(\theta))^2 + (r(\theta))^2} d\theta$$